

The Architecture of Global Air Transportation: Structural Organization Beyond Random and Scale-Free Models

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(Dated: February 3, 2026)

The global air transportation network represents a critical infrastructure system shaped by geographic, economic, and operational constraints. We present a comprehensive complex network analysis of 3,195 airports and 18,835 direct connections, characterizing structural properties including basic descriptors, clustering, path lengths, degree statistics, centrality rankings, and community organization. Degree distributions are found to follow a log-normal pattern rather than power law or exponential models. The network exhibits strong clustering ($C = 0.49$) and pronounced modularity ($Q = 0.60$), with 42 distinct communities aligned with geographic and economic regions. Comparison with null models—ten Erdős-Rényi and ten Barabási-Albert networks—reveals fundamental differences: while synthetic models reproduce sparsity and path lengths, they severely underestimate clustering and modularity, falling short of real network organization. Centrality measures show significant but non-complete overlap, indicating multi-dimensional importance in hub airports. These findings underscore the inadequacy of simple generative mechanisms and point to more complex processes involving local optimization, capacity constraints, and historic alliances. The work demonstrates that real-world transportation infrastructure exhibits organizational principles beyond classic network models, motivating future investigations into weighted, temporal, and multi-layered representations.

Keywords: airport network, complex networks, complex systems

I. INTRODUCTION

The global airline transportation system is a prototypical example of a complex network, composed of thousands of airports as nodes and direct flights as links. Representing this infrastructure as a network allows the use of mathematical and computational tools to characterize structural properties, uncover patterns, and predict system-level behaviors. Such network-based approaches have become foundational for understanding resilience, efficiency, and dynamics in transportation and infrastructure systems [1–4].

Network science has shown that real-world transportation networks are often highly heterogeneous, featuring a small number of hub airports with extensive connectivity and a large number of peripheral nodes with limited links [1, 3]. Prior studies have explored degree distributions, clustering, shortest-paths, centrality roles, and community structure in air transport networks, using both empirical data and synthetic models like Erdős-Rényi (ER) and Barabási-Albert (BA) networks to benchmark real findings [2, 4]. These methods have gained traction not only for transport but also for power grids, biological, and social systems.

In this work, we focus strictly on current air traffic data and conduct a comprehensive analysis of the main network descriptors: clustering, distances, degree statistics, centrality, communities, and we compare with

theoretical null models. Some of the main descriptors can be viewed on our website [5].

Our study aims to answer the following questions:

1. How do the empirical descriptors of the real network compare to classic random and scale-free models?
2. What is the statistical nature of the degree distribution and is it well described by standard models?
3. Which airports emerge as structurally central according to different centrality measures and do they coincide?
4. Is there clear evidence of meaningful community structure, and if so, how is it organized?

II. METHODS

A. Data Collection and Processing

The raw data for airport networks was obtained from the OpenFlights platform [6]. The dataset provides a list of airports (nodes) and direct flight connections (links) for the global air transportation system. After preprocessing—including symmetrisation—all connections in the network are bidirectional, and the network is treated as undirected (symmetric). All results except for the number of nodes, links, density, and average clustering coefficient were calculated on the largest

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connected subnetwork, which is nearly identical in size to the full network (full network: 3195 nodes, 18835 links; giant component: 3188 nodes, 18833 links). Details of data cleaning and preprocessing, including symmetrisation and giant component extraction, are documented in Appendix A.

B. Network Construction

The network is defined as a simple, undirected network where each node represents an airport and each link represents a direct connection between two airports. Multiple airlines operating the same route and parallel links are merged.

C. Software and Computational Approach

All analyses, computations, and visualizations were performed using the **NetworkX** and **Plotly** libraries in Python [7, 8]. Visualization settings are provided in the repository [9].

D. Metrics and Models

The following network metrics and models are defined and used throughout this work:

- **Number of nodes (N):** Total count of airports.
- **Degree (L):** Total count of direct flight connections.
- **Network density (D):** Defined as $D = \frac{2L}{N(N-1)}$, quantifies how many links are present compared to the maximum possible in a simple undirected network.
- **Average clustering coefficient (C):** Probability that two neighbors of a node are connected, averaged over all nodes.
- **Average shortest-path length ($\langle d \rangle$):** Average minimum number of links needed to travel between all pairs of nodes in the largest connected component.
- **Effective diameter:** 90th percentile of shortest-path distances.
- **Degree statistics:** Probability density function (PDF) and complementary cumulative distribution function (CCDF) of node degrees. Candidate fits (power law, exponential, log-normal) use formulas:

$$\text{Power law: } P(k) \sim k^{-\alpha}$$

$$\text{Exponential: } P(k) \sim e^{-\lambda k}$$

$$\text{Log-normal: } P(k) \sim \frac{1}{k} \exp\left(-\frac{(\ln k - \mu)^2}{2\sigma^2}\right)$$

- **Centrality measures:** For each node, compute and rank the top 25 by:
 - *Degree centrality:* Number of links.
 - *Closeness centrality:* Reciprocal of average shortest path length from a node to all others.
 - *Betweenness centrality:* Fraction of shortest paths passing through a node.
 - *Eigenvector centrality:* Importance of a node based on connections to other important nodes.
 - *Katz centrality:* Generalization of eigenvector centrality accounting for all paths in the network.
 - *PageRank:* Probability that a random walker lands on a node.
 - *Subgraph centrality:* Counts closed walks of all lengths originating at a node.
- **Null Models:** For structural comparison, we generated 10 Erdős-Rényi (ER) networks (N nodes, L links, probability p) and 10 Barabási-Albert (BA) networks (preferential attachment to existing nodes), both matched to the real network's size. Metrics described above are averaged over these ensembles and compared to the empirical network.
- **Community detection:** Modularity-based greedy optimization yields the best partition (Q), where

$$Q = \frac{1}{2L} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2L} \right] \delta(c_i, c_j)$$

with A_{ij} adjacency, k_i degree, c_i community, and δ the Kronecker delta.

All calculated values and model fits are reported for the real network and null models, with full documentation of all choices and procedures in the GitHub[9].

III. RESULTS

A. Network Construction and Basic Descriptors

The air transportation network comprises 3195 airports (nodes) and 18835 direct connections (links), visualized in Figure 1. Node sizes are proportional to

degree, highlighting hub airports with the highest connectivity. An interactive version, along with detailed degree statistics for every airport, is provided online [5].

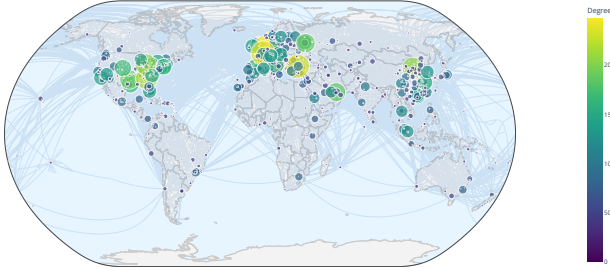


Figure 1. Global airport network: node size encodes degree.

The network density is low (0.003691), indicating a sparse structure typical of large transportation systems. The average clustering coefficient is remarkably high (0.491750), reflecting the prevalence of tightly interconnected groups of airports.

B. Properties of the Giant Component

To ensure robust analysis of path-based metrics, all results beyond basic descriptors use the largest connected component (giant component), totaling 3188 nodes and 18833 links. This subnetwork maintains the full system's sparsity (density 0.003707) and high clustering (0.492830). The mean shortest-path length is 3.96, and the effective diameter (90th percentile path length) is 5.0, quantifying strong global reach despite local clustering.

C. Degree Distribution and Model Fitting

Degree statistics for airports in the giant component reveal strong heterogeneity (min 1, max 248, mean 11.82, std 25.12). Two-thirds of airports have degree below 5, while a handful serve as major hubs (see Table I and Figure 2).

	Min	Max	Mean	Median	Std. Dev.	Nodes
Degree	1	248	11.82	3.0	25.12	3188

Table I. Basic degree statistics for airports in the giant component.

Three candidate distributions were fit to the CCDF (power law, exponential, log-normal). The log-normal distribution provided by far the best fit ($R^2 = 0.99$), indicating a highly heterogeneous, right-skewed structure. See Table II for fitting results.

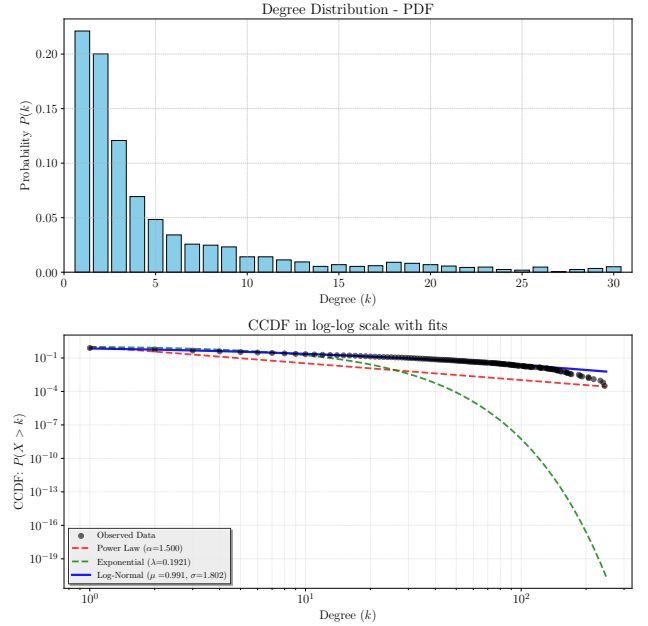


Figure 2. Degree PDF (up) and CCDF (down) for the giant component. Fits: red (power law), green (exponential), blue (log-normal).

Distribution	R^2	KS statistic
Power Law	0.36	0.27
Exponential	0.63	0.25
Log-Normal	0.99	0.07

Table II. Goodness-of-fit statistics for degree CCDF (giant component).

D. Centrality Structure

For all major centrality metrics (degree, closeness, betweenness, eigenvector, Katz, PageRank, subgraph), the top airports were ranked and compared. Overlap between rankings reveals the multi-dimensional nature of centrality in the network (see Figure 3). Full ranking tables and searchable dataframes are available online [5].

E. Null Models: Erdős–Rényi and Barabási–Albert

We generated ensembles of ten random (ER) and ten scale-free (BA) networks, matching the real system in node and link count. Table III juxtaposes mean and standard deviation of key metrics.

Degree distributions for null models are shown in Figures 4 and 5, with corresponding fits Tables IV and V. Unlike the real network, the ER model is best described

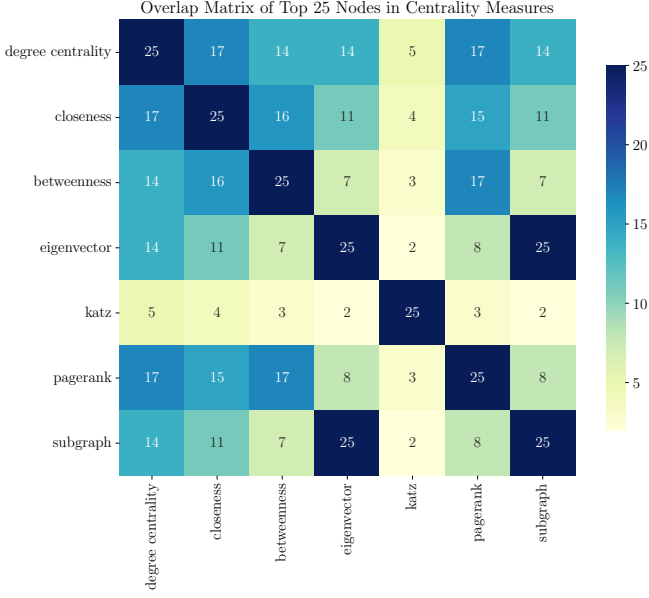


Figure 3. Overlap matrix of top 25 airports ranked by major centrality measures.

	Real	ER (avg)	BA (avg)
Density	0.003707	0.003703	0.003133
Clustering	0.492830	0.003665	0.017073
Avg path length	3.96	3.20	3.34
Degree std	25.12	3.42	11.70

Table III. Key metrics for giant component vs. null model averages.

by a log-normal, while the BA ensemble shows a dominant power-law component only in the average.

Distribution	R^2	KS statistic
Power Law	-0.81	0.99
Exponential	0.82	0.32
Log-Normal	0.99	0.03

Table IV. Goodness-of-fit statistics for degree ER.

F. Community Structure

Louvain modularity optimization detected 42 communities within the giant component ($Q = 0.60$), in stark contrast to the null models (ER: $Q = 0.24$, BA: $Q = 0.28$). Community sizes and top hubs within each group are provided in Table VI. Community visualization shown in Figure 6; for further exploration see in the interactive platform [5].

Distribution	R^2	KS statistic
Power Law	0.99	0.01
Exponential	0.97	0.07
Log-Normal	0.97	0.18

Table V. Goodness-of-fit statistics for degree BA.

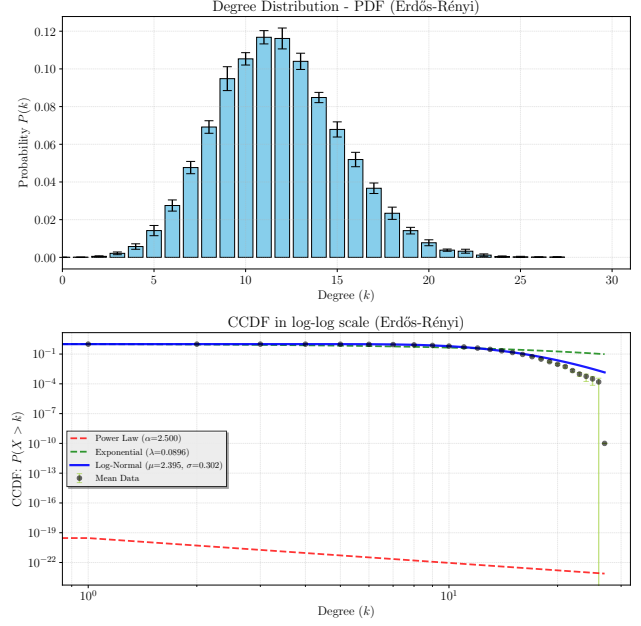


Figure 4. Degree PDF and CCDF for Erdős-Rényi (ER) model network.

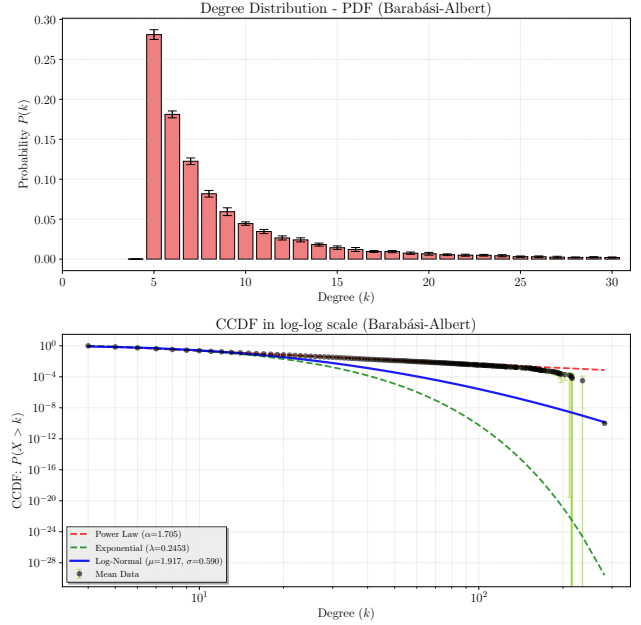


Figure 5. Degree PDF and CCDF for Barabási-Albert (BA) model network.

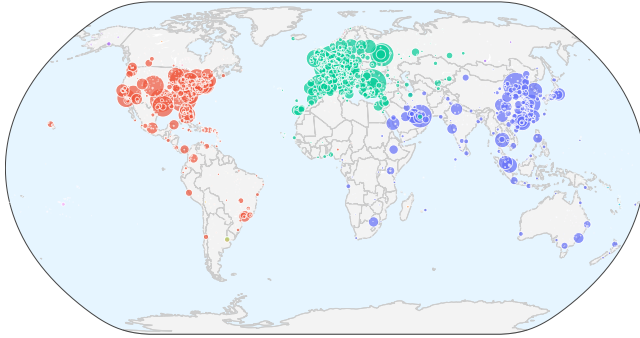


Figure 6. Detected communities in the real airport network; colours indicate community membership.

Network	Communities	Modularity
Real	42	0.60
ER	10	0.24
BA	12	0.28

Table VI. Number of detected communities and modularity (Q) scores for each network.

Group sizes range from over 900 nodes (major continental or regional blocks) down to small clusters of local airports. The modularity score in the empirical network confirms the strong presence of well-defined regional groupings a property not observed in synthetic models.

Full numerical results, interactive visualizations, and complete tables are available for exploration on the project website [5, 9]. Additionally, further plots and calculations are provided in Appendix B.

IV. DISCUSSION

The analysis of the global airport network yields clear signatures of a highly structured, non-random system. The network exhibits a sparse, yet strongly clustered topology, with most airports having few connections and a minority serving as major international hubs. Hub airports stand out not only in degree but across various centrality measures, often acting as bridges between distinct communities.

Comparing empirical results with null models underscores the limitations of classic random and scale-free constructions. While both ER and BA models can replicate the low density and typical path lengths observed in the real network, they fail to account for the high clustering coefficient and rich modular structure present in the empirical data. In particular, the real network’s modularity score ($Q = 0.60$) and numerous

small-to-medium communities (42 groups) greatly exceed those found in either ER ($Q = 0.24$, 10 groups) or BA ($Q = 0.28$, 12 groups) null models. This gap demonstrates that neither uniform randomness nor simple preferential attachment are sufficient generative mechanisms for air transport networks.

In terms of degree statistics, the real network’s distribution is best captured by a log-normal fit, rather than a pure power law or exponential model. This points to generative processes involving multiplicative growth, capacity constraints, and local optimization—consistent with the economic and regulatory complexities of the aviation sector.

Centrality rankings highlight significant overlap among high-degree, betweenness, and PageRank nodes, but also reveal unique positions for certain airports depending on the metric. Not all hubs are equally critical for connectivity or dynamics.

Community structure aligns with geographic, economic, and continental groupings, indicating that cultural/regional factors and historic airline alliances shape the effective modular organization. The null models, by contrast, generate weak or diffuse partitions without clear regional or functional coherence.

Several limitations arise from the dataset and methods. The analysis uses a snapshot of the network and does not capture temporal dynamics or seasonal fluctuations. Route frequency and passenger volume are ignored, as all links are currently treated as binary; incorporating link weights using flight counts or airline diversity would yield a richer understanding. The cleaning process may omit low-traffic airports due to missing or erroneous data.

Promising extensions include constructing weighted networks where link strength reflects flight frequency or seat capacity, and comparing network evolution over recent years to identify structural shifts or responses to global events. Examining the role of multi-layered airline alliances can deepen contextual understanding.

V. CONCLUSIONS

The structural analysis of the global airport network demonstrates that real-world transportation networks are shaped by intricate geographic, economic, and operational constraints. Unlike classic random or scale-free models, the empirical network exhibits exceptionally high clustering, striking degree heterogeneity, and a rich, modular community structure. While ER and BA null models can reproduce basic features such as sparsity and typical path lengths, they fall short in mirroring the observed modularity and clustering, underscoring the limits of simple generative rules for capturing transportation infrastructure.

Degree distributions and centrality rankings further confirm that the network’s architecture reflects more than uniform randomness or preferential attachment; its organization instead points to processes involving multiplicative growth, local optimization, and historic alliances. Meaningful community structure aligns with regional and continental zones, collectively enabling ef-

ficient connectivity and resilience. These findings highlight both the strengths and limitations of theoretical models, calling for more nuanced frameworks that incorporate empirical constraints. Future research incorporating dynamic and weighted representations will be critical for advancing our understanding of large-scale transportation networks.

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Appendix A: Data Processing

All data sources, code, and processed files for this work are publicly available in the GitHub repository **Global-Airport-Network-Analysis** [9]. The raw airport and route data were downloaded from OpenFlights [6]. The dataset contains a list of airports (nodes) and flight routes (links), which required significant cleaning prior to analysis.

First, all entries with erroneous fields (such as \N) were removed to guarantee the integrity of node and link definitions. We then constructed the network by cross-referencing routes and airports, creating a link for each valid direct connection. As the original data is directed (listing flights from airport A to B and vice versa separately), we performed symmetrisation to merge all reciprocal links, yielding an undirected (symmetric) network for analysis.

In the raw route list, each airline operating a route created a duplicate link between the same airport pair. We consolidated these multiple entries into a single undirected link, maintaining only one link per pair of airports regardless of the number of operating airlines. Any links leading to airports absent from the confirmed list due to erroneous or non-numeric IDs (“ghost” nodes) were removed, as such nodes also tend to have low degree and negligible structural relevance.

Numeric consistency was ensured by converting all

Distribution	R^2	KS statistic	$k_{\min}$
Power Law	0.232	0.746	2.9
Exponential	0.863	0.263	3.0
Log-Normal	0.999	0.029	2.0

Table VII. Goodness-of-fit statistics for degree CCDF in single ER network realization.

airport IDs within routes to numeric format and discarding any route entries for which this conversion failed. Additionally, routes beginning and ending at the same airport (“self-loops”) were excluded from the final network.

The final passes of cleaning and validation were performed within the analysis notebook, specifically removing links to missing airports. These nodes, generally of low degree (five links or fewer), do not affect the main connected structure. After all cleaning steps, the final network contained significantly fewer links than the original collection (which included roughly 67,000 directed routes from all airline companies), primarily as a result of symmetrisation and the merging of duplicate links.

Comprehensive details of all processing decisions, original table formats, and reproducible scripts are documented in the supporting materials provided on GitHub [9].

Appendix B: Supplementary Information

The following figures and numerical results complement the main analysis. Example realizations of Erdős-Rényi (ER) and Barabási-Albert (BA) networks are included, with their respective degree distributions, complementary cumulative distribution function (CCDF) fits, and key metrics (see Figures 7 and 8). Results for the degree PDF and CCDF for individual network realization are reported below, along with tables summarizing the goodness of fit for power law, exponential, and log-normal models for both ER and BA ensembles.

Table VII reports the fit statistics for the single ER network realization, while Table VIII presents the re-

sults for a single BA network realization.

Distribution	R^2	KS statistic	$k_{\min}$
Power Law	0.906	0.273	5.0
Exponential	0.827	0.273	5.0
Log-Normal	0.976	0.096	5.0

Table VIII. Goodness-of-fit statistics for degree CCDF in single BA network realization.

Additionally, we computed the correlation between each pair of centrality measures (degree, closeness, betweenness, eigenvector, Katz, PageRank, and subgraph centrality) over all node rankings. These results are visualized as a heatmap in Figure 9, illustrating which metrics identify similar sets of important nodes and highlighting subtle differences in network structure.

Further plots, extended data tables, and the complete computational workflow are available in the public repository [9] for reproducibility and deeper exploration.

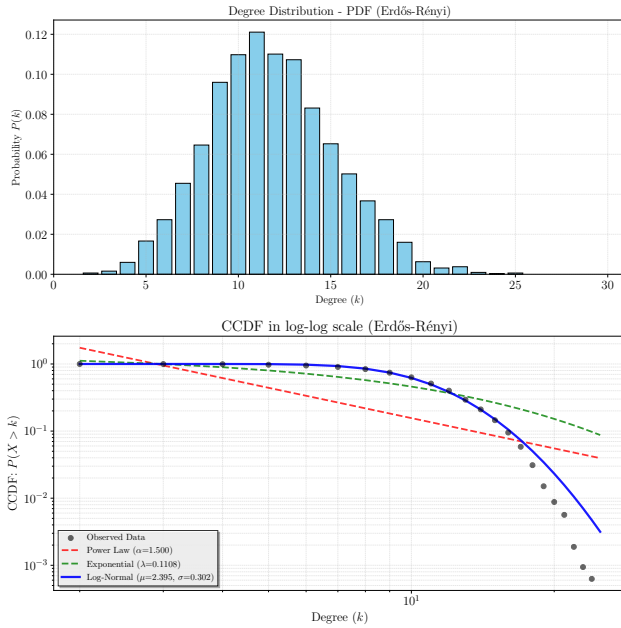


Figure 7. Degree distribution and CCDF fits for a single ER network realization.

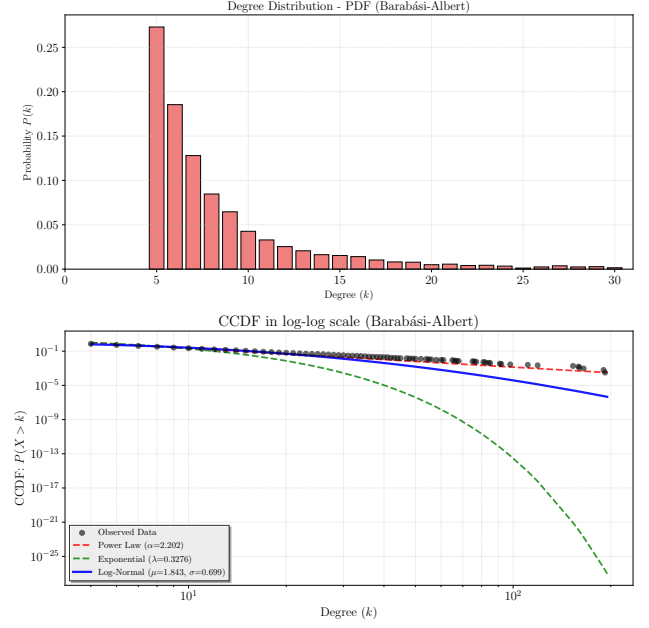


Figure 8. Degree distribution and CCDF fits for a single BA network realization.

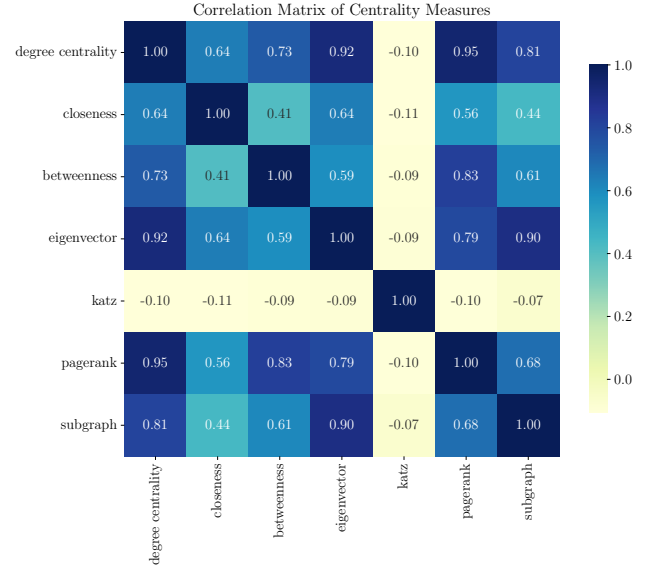


Figure 9. Correlation heatmap between centrality measures for the real network.